

Yiyang Huang

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EDUCATION

Northeastern University, Boston, USA <i>Ph.D. in Computer Engineering, advised by Prof. Yun Raymond Fu</i>	Sep 2024 – Present
Xidian University, Xi'an, China <i>M.S. in Computer Science, advised by Prof. Xuefeng Liang</i>	Sep 2021 – Jun 2024
Xidian University, Xi'an, China <i>B.Eng. in Intelligence Science and Technology</i>	Sep 2017 – Jun 2021

RESEARCH INTERESTS

Multimodal LLMs; Efficiency; Reliability; Hallucination Detection & Mitigation; Video Understanding; Layout Understanding

EXPERIENCE

Northeastern University, Boston, USA <i>SMILE Lab, Research Assistant Supervisor: Prof. Yun Raymond Fu</i>	Sep 2024 – Present
- Published 1 ACL 2026 survey paper on Video-LLM hallucinations; conducted a comprehensive review and built a curated GitHub repository of related papers and benchmarks. [GitHub] - Published 1 ICLR 2026 paper on mitigating hallucinations in MLLMs via encoder-level inference intervention. - Identified encoder-side causes of hallucinations and proposed SHIELD, a training-free module (token reweighting/subtraction + adversarial contrastive decoding), reducing hallucinations by 27% on captioning across MLLM families. - Published 1 EMNLP 2025 paper on scaling image-pretrained MLLMs to video understanding. - Developed D-CoDe, a plug-and-play framework that scales image-pretrained MLLMs to VideoQA via question decomposition and dynamic token/frame compression, improving first-person video understanding accuracy by 29% under LLaVA-NeXT.	
Adobe Research, San Jose, USA <i>Vision-Language Lab, Research Intern Mentors: Zhaowen Wang, Simon Jenni, Jing Shi</i>	May 2025 – Nov 2025
- Published 1 ECCV paper and filed 1 patent on compositional layout understanding in MLLMs. - Identified semantic drift and structural ambiguity as key challenges in compositional layout reasoning, and developed MASON with metadata-visual grounding and 3D structural perception, improving compositional matching by 26% over Gemini-2.5-Pro and 15% over GPT-5/GPT-o3 under a Qwen2.5-VL backbone. - Built CoDeLayout (~20K samples) through an automated pipeline for compositional layout mining and QA generation.	
Xidian University, Xi'an, China <i>Research Assistant Supervisor: Prof. Xuefeng Liang</i>	Sep 2021 – Jun 2024
- Published 1 ICASSP 2025 paper on visual speech recognition in low-resource language settings. - Developed a dynamic LoRA framework to learn meta lip representations shared across languages, improving accuracy by 15% on low-resource languages (e.g., Portuguese and Italian) using pretrained 3DCNN backbones and the multilingual LLM. - Published 1 ACM MM 2021 paper on contrastive and attribute learning for visual speech recognition. - Proposed CALLip, integrating attribute learning with audio-visual contrastive learning to normalize cross-speaker lip shape variations, reducing word error rate by 50% on multilingual benchmarks.	

PUBLICATIONS

Accepted

- Yiyang Huang**, Zhaowen Wang, Simon Jenni, Jing Shi, Yitian Zhang, Yizhou Wang, Yun Fu. “Beyond Atomic Layouts: Compositional Design Understanding with Vision-Language Models.” ECCV 2026.
- Yiyang Huang**, Yitian Zhang, Yizhou Wang, Mingyuan Zhang, Liang Shi, Huimin Zeng, Yun Fu. “Distorted or Fabricated? A Survey on Hallucination in Video LLMs.” ACL 2026 (Findings). [\[Paper\]](#) [\[Repo\]](#)
- Yiyang Huang**, Liang Shi, Yitian Zhang, Yi Xu, Yun Fu. “SHIELD: Suppressing Hallucinations in LVLM Encoders via Bias and Vulnerability Defense.” ICLR 2026. [\[Paper\]](#) [\[Code\]](#)
- Yiyang Huang**, Yizhou Wang, Yun Fu. “D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression and Question Decomposition.” EMNLP 2025 (Main). [\[Paper\]](#) [\[Code\]](#)
- Yiyang Huang**, Xuefeng Liang, Chaowei Fang. “CALLip: Lipreading using Contrastive and Attribute Learning.” ACM MM 2021. [\[Paper\]](#)
- Liang Shi, **Yiyang Huang**, Yun Fu. “Capturing Individual Differences of Facial Expression for Authentic Expression Generation.” IEEE FG 2026.

- Shuai Zou, Xuefeng Liang, **Yiyang Huang**. “LipReading for Low-resource Languages by Language Dynamic LoRA.” ICASSP 2025. [\[Paper\]](#)

Preprint / Under Review

- Mingyuan Zhang, Yue Bai, Yifan Wang, **Yiyang Huang**, Yun Fu. “Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models.” preprint. [\[Paper\]](#)
- Hailing Wang, Jianglin Lu, Xu Ma, **Yiyang Huang**, Yun Fu. “StreamVLM: Adapting Offline Vision-Language Models for Streaming Understanding.” NeurIPS under-review.
- Huimin Zeng, Yue Bai, Qihua Dong, Hailing Wang, **Yiyang Huang**, Yun Fu. “Narrative3D: Asset-to-Storyworld Generation.” NeurIPS under-review.

ACADEMIC SERVICE

Conference Reviewing: NeurIPS; ACL Rolling Review (ARR); IEEE Face & Gesture (FG)

Journal Reviewing: ACM TKDD

TEACHING

Teaching Assistant

DS 5110: Essentials of Data Science

Fall 2025

DS 5020: Fundamentals of Linear Algebra and Probability

Spring 2026

SKILLS

Languages: Python, $\LaTeX$

Frameworks/Tools: PyTorch, HF Transformers, HF Datasets, ffmpeg, PyAV, decord, Git, Slurm

Training/Finetuning: Accelerate, PEFT, LoRA/QLoRA, bitsandbytes

Inference: vLLM, FlashAttention

Evaluation: Imms-eval, POPE, CHAIR, MME

AWARDS

National Scholarship, China

2021

Outstanding Student, Xidian University

2022

1st Prize, Undergraduate Computer Design Competition (National Level), China

2021

2nd Prize, RoboMaster National Robotics Competition, China

2019

3rd Prize, ICRA AI Challenge

2019